Lecture 9 - Performance Evaluation

INT104 Artificial Intelligence

Evaluating Classification Methods

- Performance (for Binary classification)
  - classifier performance: predicting class label compared with groundtruth
  - True Positive (TP), True Negative (TN), False Positive (FP),False Negative (FN)
- Complexity
  - Time to construct the model (training time)
    - must be tolerable
    - can be large (lengthy)
  - Time to use the model (inference time)
    - must be tolerable
    - need for good data structures
  - Operations to train / use the model
  - Number of parameters in the models
- Robustness
  - handling noise and missing values
  - handling incorrect training data

Evaluating Classification Methods: Application Examples

1. Performance

📧 Scenario: Spam Detection

TP (True Positive):
✓ Intercepted actual "Casino Ad"

TN (True Negative):
✓ Allowed "Meeting Invite" thru

FP (False Positive):
❌ Sent boss's contract to spam

FN (False Negative):
❌ Missed a "Phishing" email

2. Complexity

🚗 Scenario: Auto-Driving

Training Time:
Supercomputers train for weeks.
👉 (Offline, lengthy is tolerable)

Inference Time:
Chips must decide in milliseconds.
👉 (Life-or-death, must be fast)

Model Parameters:
Must be compact enough to fit limited onboard memory.

3. Robustness

🏭 Scenario: Factory Sensors

Handle Noise & Missing Data:
System doesn't crash when network drops or sensor reads "9999°C". It smoothly filters anomalies.

Handle Incorrect Data:
Tired workers mislabel some machines. The model still learns correct rules despite dirty data.

Confusion Matrix

- TP, FP, FN and TN form a matrix named confusion matrix. There aremany parameters can be calculated according to the confusion matrix,which evaluate the resulting system from different perspectives.

<table>
<tr><td></td><td></td><td colspan="2">**True condition**</td><td></td><td colspan="2"></td></tr>
<tr><td></td><td>Total population</td><td>Condition positive</td><td>Condition negative</td><td>Prevalence $= \frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$</td><td colspan="2">Accuracy (ACC) $= \frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$</td></tr>
<tr><td rowspan="2">**Predicted condition**</td><td>Predicted condition positive</td><td>**True positive**</td><td>**False positive**,
Type I error</td><td>Positive predictive value (PPV), Precision $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$</td><td colspan="2">False discovery rate (FDR) $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$</td></tr>
<tr><td>Predicted condition negative</td><td>**False negative**,
Type II error</td><td>**True negative**</td><td>False omission rate (FOR) $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$</td><td colspan="2">Negative predictive value (NPV) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$</td></tr>
<tr><td></td><td></td><td>True positive rate (TPR), Recall, Sensitivity, probability of detection, Power $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$</td><td>False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$</td><td>Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$</td><td rowspan="2">Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR−}}$</td><td rowspan="2">$F_1$ score $= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$</td></tr>
<tr><td></td><td></td><td>False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$</td><td>Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$</td><td>Negative likelihood ratio (LR−) $= \frac{\text{FNR}}{\text{TNR}}$</td></tr>
</table>

Confusion Matrix

Metric	Full Name	Definition
Recall (Sensitivity)	True Positive Rate	(1)
Fall-out	False Positive Rate	(3)
Miss rate	False Negative Rage	(2)
Specificity (Selectivity)	True Negative Rate	(4)

$$recall = \frac{TP}{TP + FN} = \frac{TP}{n^+} \quad (1)$$

$$fallout = \frac{FP}{TN + FP} = \frac{FP}{n^-} \quad (3)$$

$$miss = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}} = \frac{\mathrm{FN}}{n^+} \quad (2)$$

$$specificity = \frac{TN}{TN + FP} = \frac{TN}{n^-} \quad (4)$$

NB: Recall is also named sensitivity

Confusion Matrix

More general measurements:

$$precision = \frac{TP}{TP + FP} \tag{5}$$

$$accuracy = \frac{\mathrm{TN} + \mathrm{TP}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} = \frac{\mathrm{TN} + \mathrm{TP}}{n} \tag{6}$$

A good measurement should balance the multiple aspects such as F_1 Score (F-Measure)

$$F_1 \mathrm{score} = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = 2 \cdot \frac{\mathrm{precision} \cdot \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{7}$$

- **Total Patients** = 100
- **Actual Condition**: 15 Diseased (Positive), 85 Healthy (Negative).
- **Model Predictions**: **TP** = 10, **FN** = 5, **FP** = 15, **TN** = 70.

Accuracy	(TP+TN) / Total	The proportion of total correct predictions. *Overall correctness.*	(10+70)/100 = 80%	Overall, the model made the correct judgment for 80% of the patients.
Precision (Positive Predictive Value)	TP / (TP+FP)	The proportion of true positive predictions out of all positive predictions. *How reliable is the alarm?*	10 / (10+15) = 40%	Out of all patients diagnosed as "diseased", only 40% actually had the disease.
Recall (Sensitivity / TPR)	TP / (TP+FN)	The proportion of true positive cases correctly identified. *How many actual cases were found?*	10 / (10+5) ≈ 66.7%	Out of the 15 actually diseased patients, the model successfully found ~67%.
Specificity (TNR)	TN / (TN+FP)	The proportion of true negative cases correctly identified. *How many healthy cases were ruled out?*	70 / (70+15) ≈ 82.4%	Out of the 85 actually healthy people, the model correctly confirmed ~82% as healthy.
Fall-out (FPR)	FP / (FP+TN)	The proportion of actual negative cases incorrectly identified as positive. Mathematically: 1 - Specificity	15 / (15+70) ≈ 17.6%	~18% of the healthy people were falsely alarmed with a positive diagnosis.
Miss Rate (FNR)	FN / (FN+TP)	The proportion of actual positive cases incorrectly identified as negative. Mathematically: 1 - Recall	5 / (5+10) ≈ 33.3%	~33% of the actual diseased patients were completely missed (fatal error in medicine).
F1-Score	2*(P*R) / (P+R)	The harmonic mean of Precision (P) and Recall (R). Used for imbalanced datasets.	2*(0.4*0.667) / (0.4+0.667) = 50%	A balanced metric showing moderate performance due to the low Precision.

Exercise

Please calculate Accuracy, Specificity, Recall and F_1 score for the givencases

TP	FP	FN	TN	Accu	Recall	Precision	F1 Score
63	28	37	72				
77	77	23	23				
24	88	76	12				
76	12	24	88				

1. Accuracy (TP + TN) / Total (63 + 72) / 200 = 135 / 200 = **0.6750**
2. Recall TP / (TP + FN) 63 / (63 + 37) = 63 / 100 = **0.6300**
3. Precision TP / (TP + FP) 63 / (63 + 28) = 63 / 91 ≈ **0.6923**
4. F1 Score 2TP / (2TP+FP+FN) (2*63) / (126+28+37) = 126/191 ≈ **0.6597**
5. Specificity TN / (TN + FP) 72 / (72 + 28) = 72 / 100 = **0.7200**

Receiver Operation Characteristics (ROC)

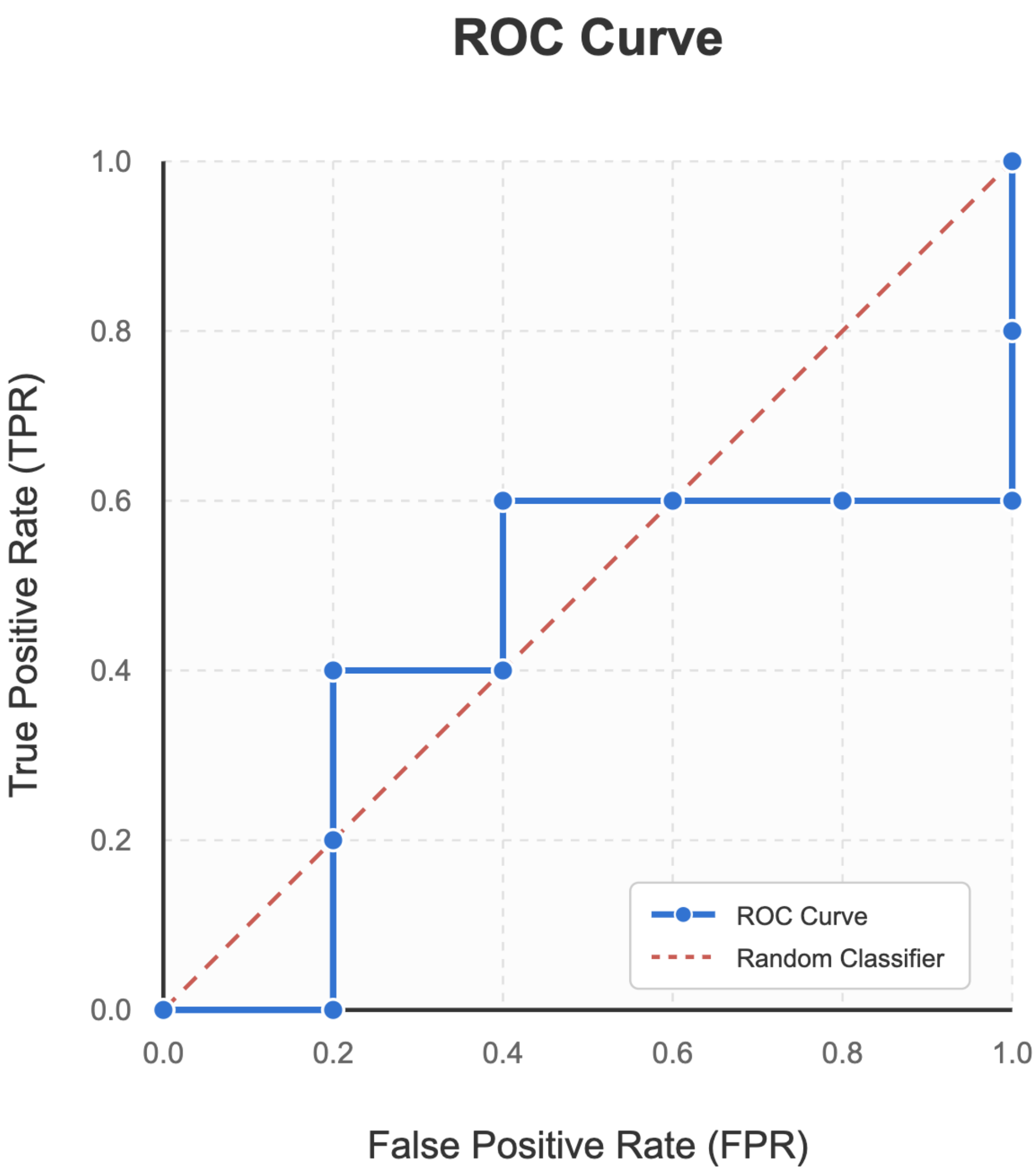
Assume that the following scores are obtained from classifiers and youwould like to set a threshold for decision making. How do you evaluatethe system performance of different threshold values?

Predict Score	Label
0.11	1
0.35	0
0.92	0
0.51	1
0.49	0
0.81	1
0.26	0
0.63	0
0.72	1
0.25	1

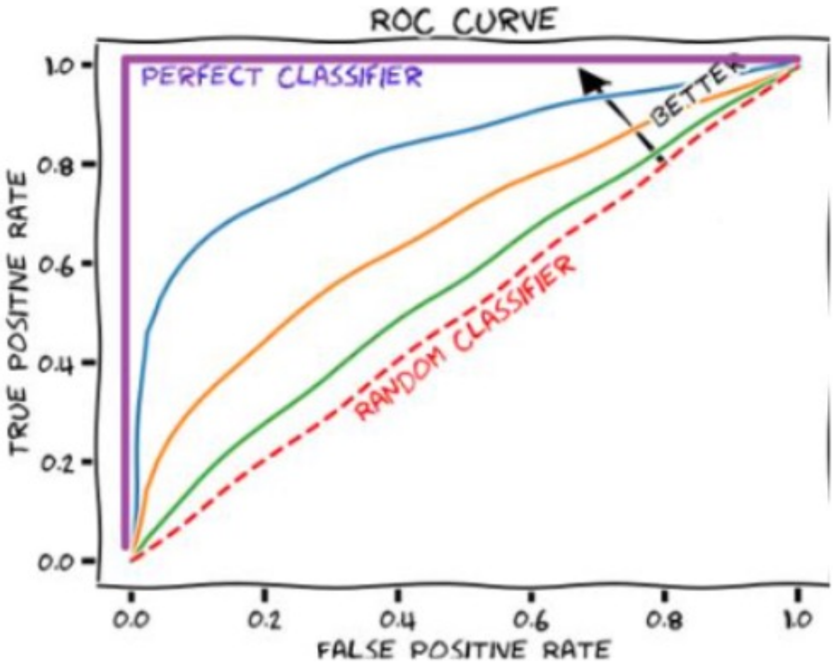
Receiver Operation Characteristics (ROC)

TPR (True Positive Rate / Recall) = TP / Actual Positives
FPR (False Positive Rate) = FP / Actual Negatives

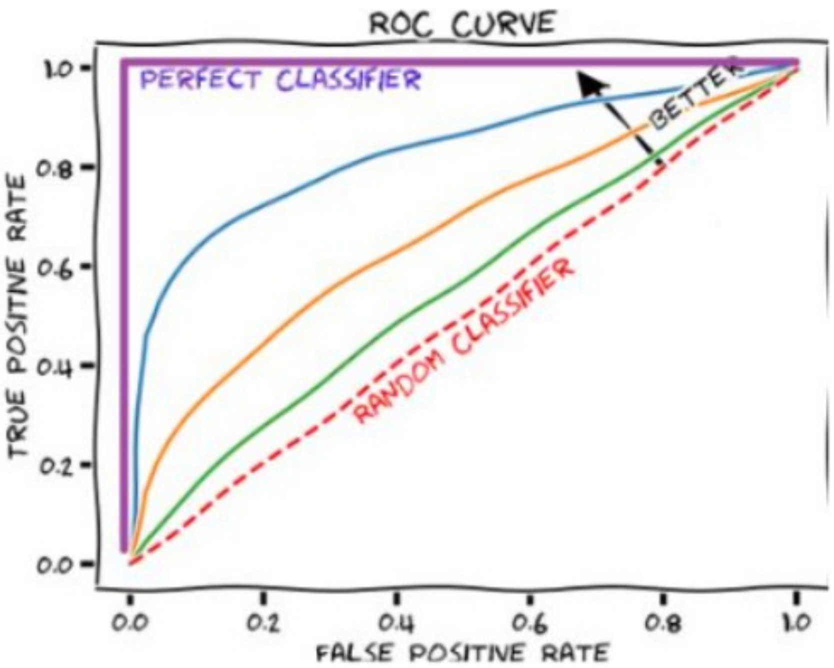
Threshold	Label	TP	FP	TPR (TP/5)	FPR (FP/5)
≥ 0.92	0	0	1	0.0	0.2
≥ 0.81	1	1	1	0.2	0.2
≥ 0.72	1	2	1	0.4	0.2
≥ 0.63	0	2	2	0.4	0.4
≥ 0.51	1	3	2	0.6	0.4
≥ 0.49	0	3	3	0.6	0.6
≥ 0.35	0	3	4	0.6	0.8
≥ 0.26	0	3	5	0.6	1.0
≥ 0.25	1	4	5	0.8	1.0
≥ 0.11	1	5	5	1.0	1.0


ROC Curve
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
ROC Curve
Random Classifier

Receiver Operation Characteristics (ROC)



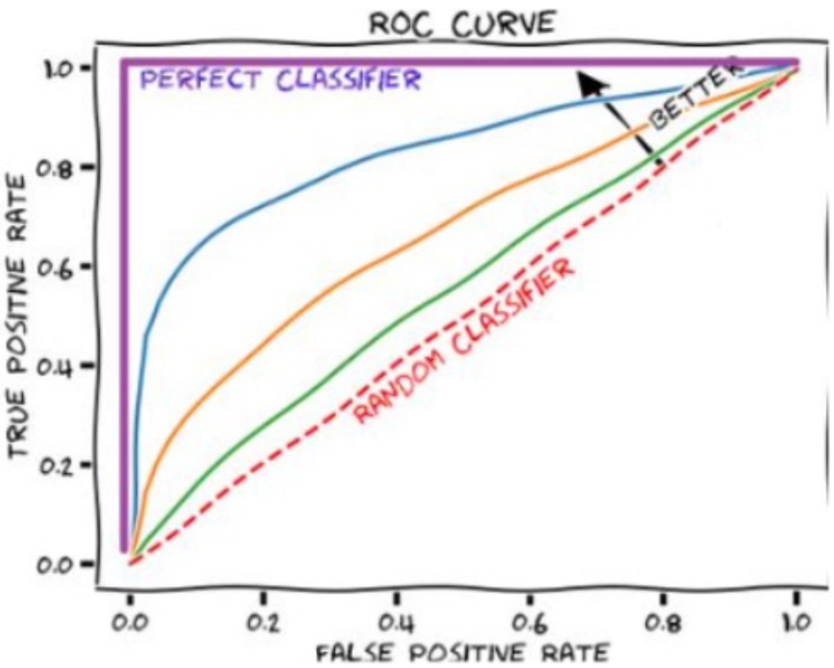
- One of the measurement for ROC test is Area under Curve (AUC)
- AUC>0.5, the system performance is better than random guess.



It's a simple way to check how good your classification model is

- Y-axis (True Positive Rate): How many *real positive cases* your model catches correctly. Higher = fewer misses.
- X-axis (False Positive Rate): How many *real negative cases* your model incorrectly flags as positive. Lower = fewer false alarms.

- The **purple line (Perfect Classifier)** 100% accurate: It catches every positive case with *zero* false alarms. (This is the ideal, almost never happens in real life!)
- The **red dashed line (Random Guess)** No better than flipping a coin. Right only half the time.
- The **blue/orange/green lines (Your models)** The closer a line gets to the top-left corner, the better the model.
- The arrow labeled "BETTER" points exactly where you want your curve to go!



AUC (Area Under the Curve)
AUC is the single number that sums up how good your ROC curve is.
The bigger the area under your curve, the better the model:

- AUC = 1.0 → Perfect (like the purple line)
- AUC = 0.5 → No better than random guess (the red dashed line)
- AUC > 0.5 → Your model is actually learning something!

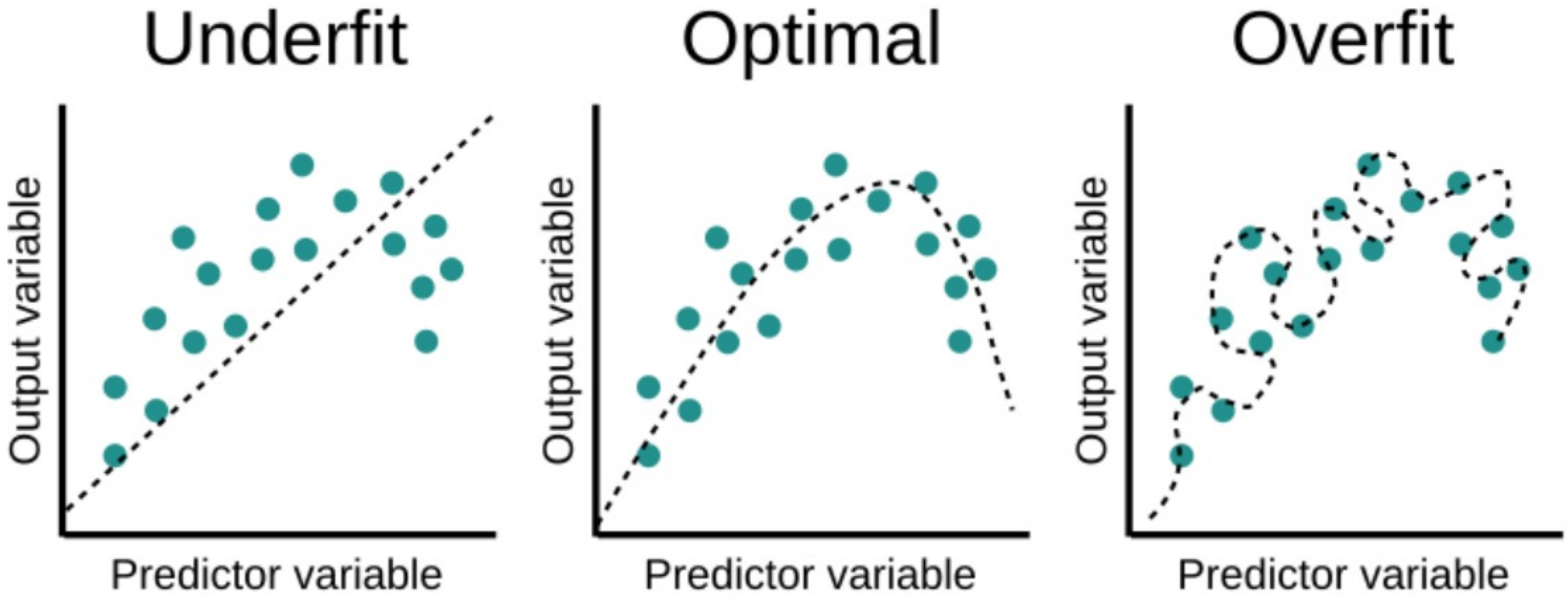
AUC can never be > 1.0! The higher it is (up to 1), the better.

Cross Validation

- Ideally, the training data should have the same distribution in practice,which never happens ...
- How do we test the performance of resulting models?
- We could use a part of training data as the validation set
- By performing cross validation, we expect the validation set must be agood representative for the whole data

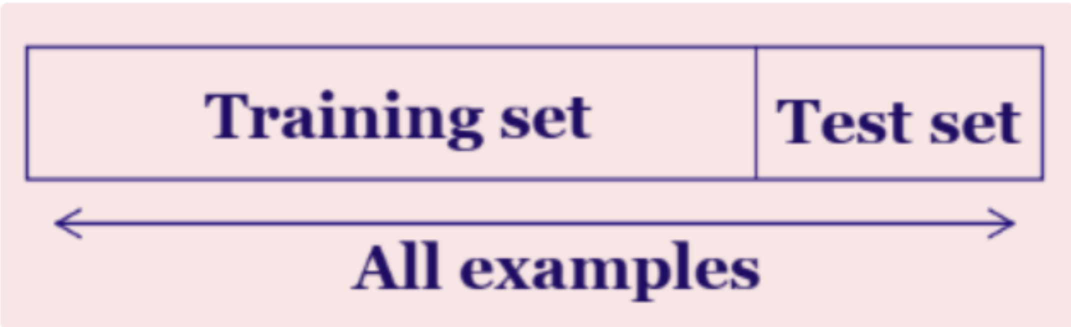
Overfit

- One of the most common objective for cross validation is to avoid overfitting
- If a system is too complex, the performance of the system could be too good to be true



Data Partitioning Methods for X-Validation

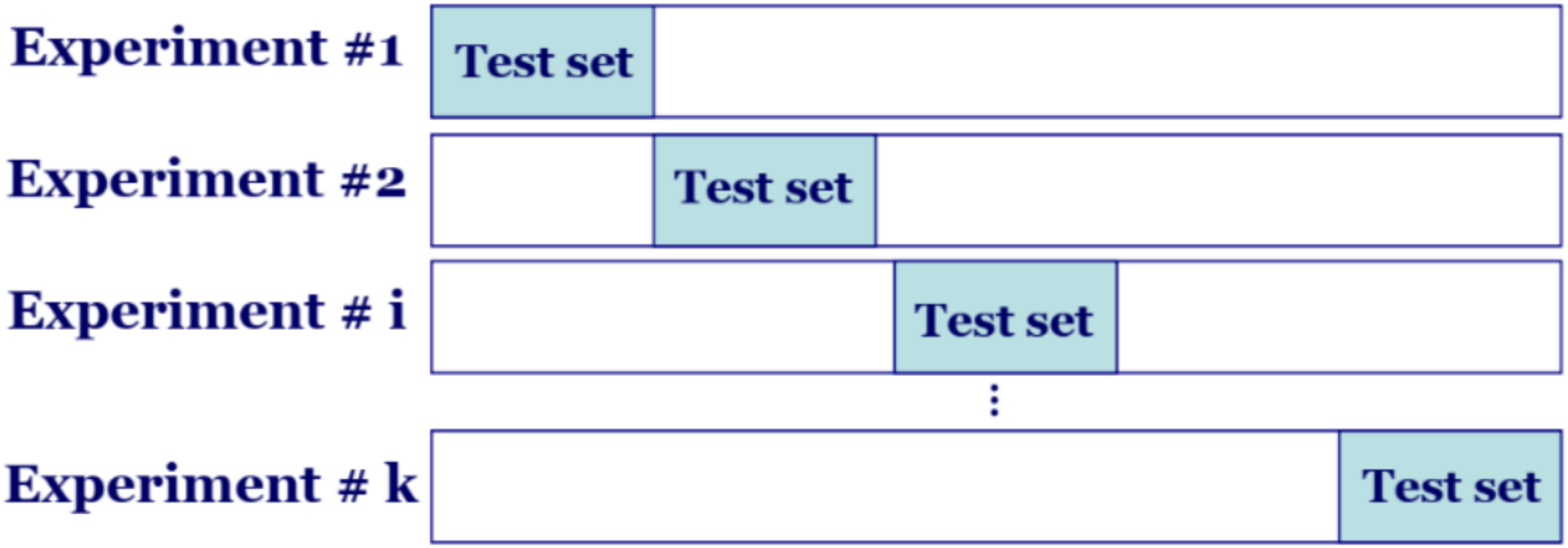
- Random Sampling
  - data is randomly partitioned into two separate sets
  - Assumption: data is uniformly distributed

- Bootstrap
  - resample with replacement n sample of original data as training set.
  - some numbers in the original sample may be included several timesin the bootstrap sample
  - can be used to estimate data distribution

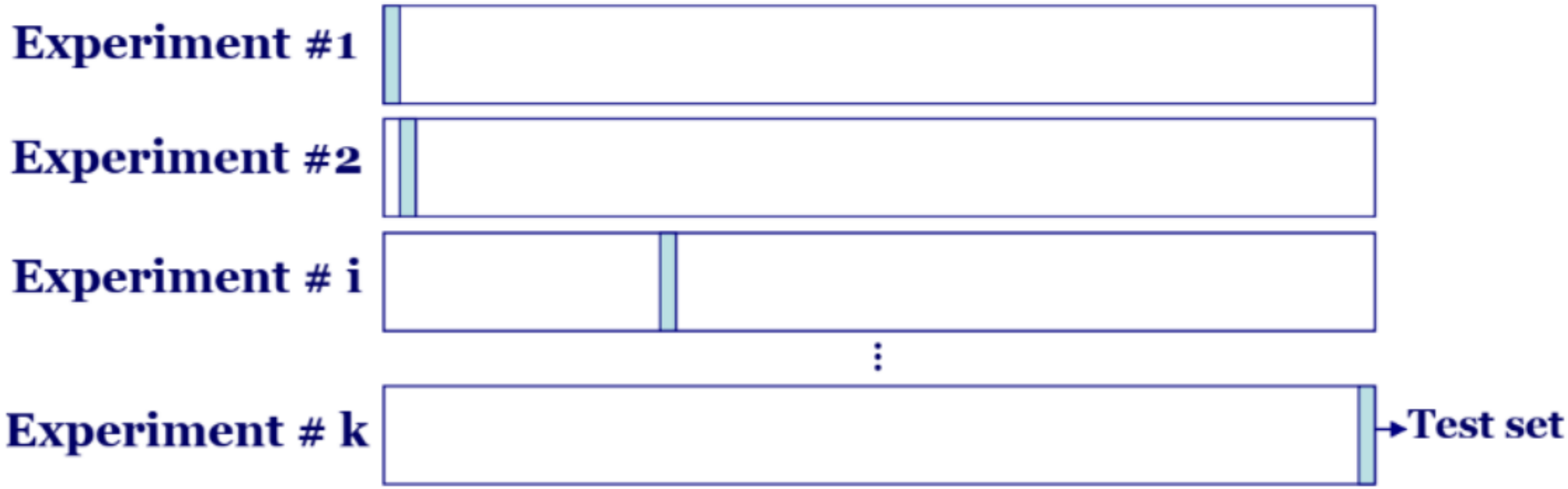
k-fold Cross Validation

- k-fold (k is usually set to 5, 10 and other numbers)
  - Randomly partition the data into k mutually exclusive subsets, eachapproximately equal size
  - At i -th iteration, use D_i as test set and others as training set
  - The mean of measures obtained in iterations used as the output ofperformance measure



Leave-one-out Cross Validation

- k folds where k is the number of samples.
- The method is particularly useful for small sized dataset.



Three-way Data Splits

- If model selection and true error estimates are to be computedsimultaneously, the data needs to be divided into three disjoint sets
  - Training set: a set of examples used for learning: to fit theparameters of the classifier
  - Validation set: a set of examples used to tune the parameters of aclassifier
  - Test set: a set of examples used only to assess the performance ofa fully-trained classifier
- Why separate test and validation sets?
  - The error rate estimate of the final model on validation data will bebiased(smaller than the true error rate) since the validation set isused to select the final model
  - After assessing the final model with the test set, YOU MUSTNOT tune the model any further

Model Selection Criteria

- For model-based method, model likelihood could be used as a metric of system performance
- Please remember:
  - the model likelihood can only be compared with the same testing dataset
  - the model likelihood should be compared in the basis of per sample
- To prevent the overfitting problem, model complexity should be considered

✍️ The typical way to evaluate a model is model selection criteria.

Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC) are

$$\mathrm{AIC} = 2k - 2\ln(\mathcal{L})$$

$$\mathrm{BIC} = k\ln(n) - 2\ln(\mathcal{L})$$

AIC and BIC

These are common **model selection criteria** that balance **fit to data** and **model complexity**:

- **AIC (Akaike's Information Criterion)**: $\mathrm{AIC} = 2k - 2\ln(\mathcal{L})$
- **BIC (Bayesian Information Criterion)**: $\mathrm{BIC} = k\ln(n) - 2\ln(\mathcal{L})$
- Where:
  - k = number of model parameters (complexity)
  - $\mathcal{L}$ = likelihood of the model
  - n = number of data samples

How to use them: Lower AIC or BIC means a better model (better fit + reasonable complexity).